Q1.

Ainee is the owner of several car shops and employs multiple managers who oversee the daily operations. She stores the data about her business in a relational database named "CARS." The database has three tables: SHOP, MANAGER, and CAR.

The SHOP table contains columns for `ShopID`, `ManagerID`, `Address`, `Town`, and `TelephoneNumber`.

The `MANAGER` table has columns for `ManagerID`, `FirstName`, `LastName`, `DateOfBirth`, and `Wage`.

The `CAR` table has columns for `RegistrationNumber`, `Make`, `Model`, `NumberOfMiles`, and `ShopID`.

a) Identify which fields in each table are primary keys and which ones are foreign keys.

Answer.

Table: SHOP

ShopID: Primary key

ManagerID: Foreign key

Table: MANAGER

ManagerID: Primary key

Table: CAR

RegistrationNumber: Primary key

ShopID: Foreign key

[2]

b) Explain how access rights can be used to safeguard the data in Ainee's database against unauthorized access.

Answer.

- **User-specific permissions**: Assign different access levels to users based on their roles.
- **Restricted access**: Limit each user to only the data they need (e.g., managers access only their shop's data).
- **Full control for Ainee**: Ainee retains full access to all database data.
- **Authentication measures**: Implement passwords and authentication protocols to prevent unauthorized access.

[3]

c) (i) Ainee uses DDL and DML statements in her database. Complete the following DML statements:

To retrieve the number of cars for sale in each shop.

```
SELECT ShopID, COUNT(*) AS NumberOfCars

FROM CAR

GROUP BY ShopID;
```

[3]

(ii) To insert a new record for a car with the following data: RegistrationNumber: 123AA, Make: Tiger, Model: Lioness, NumberOfMiles: 10500, ShopID: 12BSTREET.

```
INSERT INTO CAR (RegistrationNumber, Make, Model, NumberOfMiles, ShopID)

VALUES ('123AA', 'Tiger', 'Lioness', 10500, '12BSTREET');
```

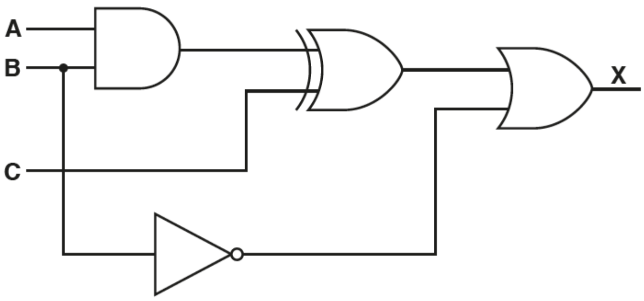
```
INSERT INTO CAR

VALUES ('123AA', 'Tiger', 'Lioness', 10500, '12BSTREET');
```

[2]

Q2.

A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

```
((A AND B) XOR C) OR NOT B
```

Zak A AND B …
Zak … XOR C …
Zak … OR NOT B

[3]

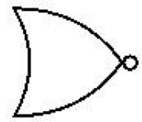
(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		0

[2]

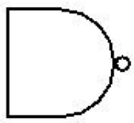
(c) Identify one logic gate not used in the given logic circuit. Draw the symbol for the logic gate and complete its truth table.

NOR



A	B	OUTPUT
0	0	1
0	1	0
1	0	0
1	1	0

NAND



A	B	OUTPUT
0	0	1
0	1	1
1	0	1
1	1	0

[3]

Q3.

(a) Dry-run this assembly language program using the trace table.

ACC	IX	A	B	C	D	300	301	302
300	0			300				
0					0			
3		3						
1			1					
3								
4						4		
0								
1					1			
	1							
4		4						
2			2					
4								
6							6	
1								
2					2			
	2							
5		5						
8			8					
5								
13								13
2								
3								

[5]

(b)

(i) What is the purpose of the given assembly language code?

Answer.

The program performs **element-wise addition of two arrays (Array1 and Array2) and stores the results in a third array (Array3).**

[4]

(ii) What is the size of the three arrays?

Answer.

Each array has 3 elements.

- Array1: 3 locations (100, 101, 102).
- Array2: 3 locations (200, 201, 202).
- Array3: 3 locations (300, 301, 302).

[1]

(iii) What is the significance of the label "loop" in the code?

Answer.

The label "LOOP:" marks the beginning of a loop that iterates over the elements of arrays 1 and 2, adding them together and storing the result in array 3.

[1]

(iv) What does the instruction "CMP #3" do?

Answer.

Purpose: Compares the contents of the accumulator (ACC) with the number **3**.

[1]

(v) How would you modify the code to subtract content of B from A and store the result in address saved in C?

Answer.

To subtract content of B from A and store the result in C, the program would need to be modified to replace the `ADD B` instruction with a `SUB B` instruction.

[1]

Q4:

(a) Denary equivalents of the following IEC binary prefixes used to represent quantities of digital information:

1 Kibi: 1024

1 Mebi: 1048576

1 Gibi: 1073741824

[3]

(b) To convert 1/2 TB to the nearest number of GiB:

1/2 TB = 500 GB

1 GiB = 2^30 bytes = 1073741824 bytes

[2]

Therefore, 500 GB is equal to approximately 465.66 GiB (rounded to two decimal places).

(c) Perform the following binary addition. Show your working.

(i)

```
    1101 0000
    1011 0000
----------------------------------
1/  1000 0000
```

[2]

(ii) Yes, overflow occurred because adding two negative numbers produced a result outside the representable range for 8-bit signed numbers.
Since the expected result is negative but the 8-bit truncated value **does not match expectations**, overflow is detected.

[1]

(iii) The result in binary is `1000 0000`, which is equal to **-128** in denary. **[1]**

(iv) The result in binary is `1000 0000`, which is equal to **80** in hexadecimal. **[1]**

Q5:

ZAK wants to protect the sensitive data on his company's computer network from unauthorized access and malware attacks.

(a) Explain the importance of data encryption and access controls in ensuring data security on a computer network.

Answer:

Data Encryption: Converts plain text into a coded format to protect sensitive information.
Authorized Access: Only users with the decryption key can read the encrypted data.
Access Controls: Restrict access to sensitive data, allowing only authorized users.
Enhanced Security: Prevents unauthorized access and reduces the risk of data breaches.

[3]

(b) ZAK is considering implementing antivirus software on his company's computers. Describe how antivirus software works to protect against malware attacks and provide an example of a common type of malware.

Answer:

How Antivirus Software Protects Against Malware:
Scans files and programs for malware signatures or patterns.
Quarantines or deletes detected malware.
Provides real-time protection by monitoring system activity.
Regular updates to detect new threats.

Example of Malware – Spyware: [Any malware is accepted.]
Disguises as harmless software.
Steals sensitive information.
Compromises system security.

[3]

(c) Two common types of cyber-attacks are phishing and pharming. Describe the difference between phishing and pharming and provide an example of each.

[2]

Answer:

Difference Between Phishing and Pharming

Phishing: Tricks users into revealing sensitive information through fake emails or websites.
    Example: An email pretending to be from a bank, asking users to enter their login credentials.

Pharming: Redirects users to a fake website, even when they enter the correct URL.
    Example: A fake banking website that looks real, stealing users' login credentials.

Q6.

Harim is starting her first job as a software developer for a multinational company.

(a) Harim has been advised to join a professional ethical body.

Explain the benefits to Harim of joining a professional ethical body.

Answer:

Benefits of Joining a Professional Ethical Body:

- **Demonstrates professionalism** and commitment to ethical standards.
- **Enhances reputation** among colleagues and employers.
- **Provides resources and guidance** on ethical issues.
- **Offers networking opportunities** with industry professionals.
- **Supports professional development** through conferences and training.

[3]

(b) Harim is shown the software she will be working on. She is unfamiliar with the Integrated

Development Environment (IDE) she is required to use.

(i) Describe the ethical ways in which Harim can act in this situation.

Answer:

Ethical Actions Harim Can Take:

- **Ask for guidance** from experienced colleagues or supervisors.
- **Research the IDE** using online resources and company documentation.
- **Seek training** or educational resources to improve her skills.
- **Be transparent** about her lack of experience.
- **Communicate concerns** or limitations to her team.

[2]

(ii) In addition to debugging tools, identify three other tools or features commonly found in an IDE that support the writing of a program.

Answer.

In addition to debugging tools, some other common tools or features found in an IDE that support the writing of a program are:

- Colour coding // pretty printing
- Auto-complete
- Auto-correct
- Context sensitive prompts
- Expand and collapse code blocks
- Code completion or auto-suggest
- Syntax highlighting

[3]

(c) Harim is part of a team writing a program. She finds an error in part of the program that

has already been tested. She decides not to tell anyone because she is worried about the consequences.

Explain the reasons why Harim acted unethically in this situation.

Answer.

Why Harim Acted Unethically:
• **Jeopardizes software quality** and functionality.
• **Potential harm to end-users** due to unreported errors.
• **Violates ethical standards** of transparency and accountability.
• **Risks legal and financial consequences** for the company.
• **Undermines teamwork** by not communicating critical issues.

[2]

<u>**Compilers**</u> are usually used when a high-level language program is complete.
They translate all the code at the same time and then run the program.
They produce <u>**executable/.exe/object code**</u> files that can be run without the source code.
<u>**Interpreters**</u> translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.
Assemblers are used to translate assembly code into <u>**binary/machine code**</u>.

[4]

Q7.

(a) Table below shows four stages in converting sound into a digital form.

Show the correct order for the stages by labelling them with the numbers 1 – 4

(1 being the first stage).

Answer.

(a)

1 microphone picks up sound waves
2 converted to an electrical analogue signal
3 values read at specific point and rounded to a level
4 Binary representation of level stored

[3]

(b) Describe how a black and white image could be represented as a bitmap in binary.

Answer:

Representation of a Black and White Image as a Bitmap:
- **Divided into a grid of pixels** to form a matrix.
- **Each pixel is assigned a binary value**:
  - 0 for black
  - 1 for white
- **Stored as a binary sequence**, with each pixel represented by a single bit.
- **Bitmap file includes File Header**, stores image dimensions and pixel data etc.

[3]

(c) A screen contains black text on a solid white background. One line of the screen might be represented as follows with W representing white pixel and B representing a black pixel.

`WWWWWWBBBWWWWWWWWWWWWBBBBBBWBBBWWWW`

A compression technique is applied to the line of data above and results in the following output:

`6W3B12W6B1W3B4W`

State what data compression algorithm has been applied.

Answer.

The compression technique that has been applied is **run-length encoding**.

[1]

(d) Another data compression technique is JPEG, which is a lossy compression algorithm.

Explain what is meant by lossy compression.

Answer.

Lossy Compression Explanation:
- **Reduces file size** by discarding some data.
- **Not an exact copy** of the original, leading to **loss of quality**.
- **Higher compression rates** compared to lossless methods.
- **Common in media formats** like **JPEG (images)** and **MP3 (audio)**.
- **Removes less noticeable details**, such as subtle color variations in images. **[1]**

Q8.

The two main types of networks are **LAN (Local Area Network)** and **WAN (Wide Area Network)**. The key difference between them is that a LAN is confined to a small geographic area, while a WAN can cover a larger geographic area, such as a city, country, or even the entire world.

The two main models of networked computers are **client-server** and **peer-to-peer**. In the client-server model, one computer (the server) provides services or resources to other computers (the clients) on the network. In the peer-to-peer model, all computers on the network are equal and can share resources with each other.

The different types of subnetwork models include bus, star, **mesh**, and **hybrid** topologies. Each has its own benefits and drawbacks, such as scalability, reliability, and ease of implementation.

The hardware used in LAN networks includes switches, servers, NICs, WNICs, WAPs, cables, bridges, and repeaters. These devices are used to connect computers and other devices on the network, and facilitate communication between them.

The role of a **router** in a network is to direct data packets between different networks. Ethernet is a set of rules and protocols that govern how data is transmitted over a LAN network, and includes mechanisms to avoid collisions, such as **Carrier Sense Multiple Access/Collision Detection (CSMA/CD).**

Cloud computing is a model for delivering on-demand computing resources over the internet. **Public clouds** are available to anyone over the internet, while **private clouds** are reserved for a single organization or business. The benefits of cloud computing include scalability, flexibility, and cost-effectiveness, while drawbacks include security and privacy concerns.

[8]